Lecture 18: Section 2.7
Nonlinear Inequalities

Solving polynomial inequalities

Key (critical) numbers

Solving rational inequalities

To Solve a Polynomial Inequality $2x^2 - 3x > 5$

1. Write the inequality with 0 on the right hand side to obtain one of the following:

$$f(x) > 0 \quad f(x) \geq 0 \quad f(x) < 0 \quad f(x) \leq 0$$

2. Find all real zeros of the polynomial $f(x)$. These zeros are the **key (critical) numbers** of $f(x)$.

3. Use the key (critical) numbers to determine the test intervals.

4. Choose a number in each interval and evaluate f at the number.

a) If the function value is positive, then $f(x) > 0$ for all numbers x in the interval.

b) If the function value is negative, then $f(x) < 0$ for all numbers x in the interval.

5. Include endpoints if $f(x) \geq 0$ or if $f(x) \leq 0$.

FACT: A polynomial changes signs only at its zeros. Between two zeros, a polynomial must be entirely positive or entirely negative.

ex. Solve the inequalities. Write your answer in interval notation.

a) $2x^2 - 3x > 5$

(1) $2x^2 - 3x - 5 > 0$

(2) $(2x-5)(x+1) > 0$

$x = \frac{5}{2}$ $\quad x = -1$

(3)

test $x=-2$ $\quad x=0$ $\quad x=3$

$- \cdot -$ $\quad - \cdot +$ $\quad + \cdot +$

(+) (−) (+)

-1 $\quad \frac{5}{2}$

$\boxed{(-\infty, -1) \cup (\frac{5}{2}, \infty)}$

b) $x^4 \leq 8x$

① $x^4 - 8x \leq 0$

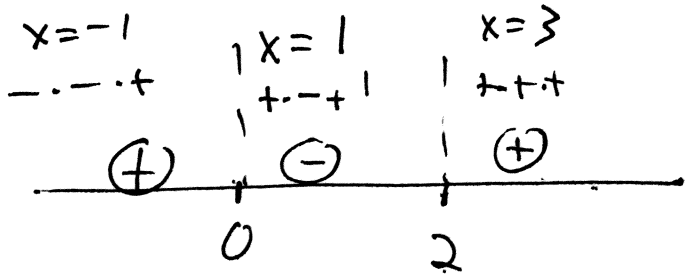
② $x(x^3 - 8) \leq 0$

$x(x^3 - 2^3) \leq 0$

$x(x-2)(x^2+2x+4) \leq 0$

③ $x=0 \quad x=2$

④



$\boxed{[0,2]}$

$x^2+2x+4=0$

$x = \dfrac{-2 \pm \sqrt{2^2-4(1)(4)}}{2(1)}$

$x = \dfrac{-2 \pm \sqrt{-12}}{2}$

no real solutions.

Don't divide by a variable.

$x^4 \leq 8x$

$x^3 \leq 8$

$x^3 - 8 \leq 0$

$(x-2)(x^2+2x+4) \leq 0$

we lost $x=0$ as a zero.

Checkpoint: Lecture 18, problem 1

To Solve a Rational Inequality $\dfrac{x^2-1}{x^2+4x+2} > 0$

1. Write the rational inequality with 0 on the right side. Write the left side as a **single** fraction. Note the domain.

2. Find all zeros of numerator and denominator. These zeros are the **key (critical) numbers** of a rational inequality.

3. Use the key (critical) numbers to determine the test intervals.

4. Solve as before, including endpoints **in the domain** if they are solutions.

FACT: The value of a rational expression can change signs at its zeros and its undefined values. These two types of numbers make up the **key (critical) numbers** of a rational inequality.

$$\frac{-x^2+3x+4}{(x-2)^2} > 0$$

ex. Solve the rational inequalities. Write your answer in interval notation.

(1) ✓

(2) ✓

a) $$\frac{(x+1)(4-x)}{(x-2)^2} > 0$$

$\geq$

$[-1$

$0 \cancel{>} 0$

(3) $x=-1$, $x=4$, $x=2$

(4)

$x=-2$	$x=0$	$x=3$	$x=5$
$\frac{-\cdot+}{+}$	$\frac{+\cdot+}{+}$	$\frac{+\cdot+}{+}$	$\frac{+\cdot-}{+}$
$-$	$+$	$+$	$-$

-1 $\quad$ 2 $\quad$ 4

undefined

$$\boxed{(-1,2)\cup(2,4)}$$

You might think you know a quicker way.

b) $\dfrac{6}{2x-5} \le 2$

Incorrect

$(2x-5)\left(\dfrac{6}{2x-5}\right) \le 2(2x-5)$

$6 \le 4x-10$

$16 \le 4x$

$4 \le x$

$x \ge 4$

$[4, \infty)$

$-4x+16=0$

$-4x = -16$

$x = 4$

① $\dfrac{6}{2x-5} - 2 \le 0$

② $\dfrac{6}{2x-5} - 2\left(\dfrac{2x-5}{2x-5}\right) \le 0$

$\dfrac{6-2(2x-5)}{2x-5} = \dfrac{6-4x+10}{2x-5} = \dfrac{-4x+16}{2x-5} \le 0$

$2x-5=0$

$2x=5$

$x=\frac{5}{2}$

$0 \le 0$ true

③ $x=0$ $-$ $\quad$ $x=3$ $+$ $\quad$ $x=5$ $-$

$-$ $\quad$ $+$ $\quad$ $-$

$\frac{5}{2}$ $\quad$ 4

und.

$\boxed{\left(-\infty, \frac{5}{2}\right) \cup [4, \infty)}$

NOTE: Do not clear fractions in an inequality when there is a variable in the denominator.

$\dfrac{6}{2x-5} \le 2$

Checkpoint: Lecture 18, problem 2

$6 \le 2(2x-5)$

$6 \le 4x-10$

$16 \le 4x$

$4 \le x$ $\quad [4, \infty)$ we lost part of the solution.

c) $-\frac{3}{x} + \frac{4}{x-2} \leq \frac{3}{x} - \frac{3}{x}$

① $\frac{4}{x-2} - \frac{3}{x} \leq 0$

② $\left(\frac{4}{x-2}\right)\frac{x}{x} - \left(\frac{3}{x}\right)\left(\frac{x-2}{x-2}\right) \leq 0$

$$\frac{4x-3(x-2)}{x(x-2)} = \frac{4x-3x+6}{x(x-2)} = \frac{x+6}{x(x-2)} \leq 0$$

③ $x=-7$ $\quad$ $x=-1$ $\quad$ $x=1$ $\quad$ $x=3$

$\frac{-}{--}$ $\quad$ $\frac{+}{--}$ $\quad$ $\frac{+}{+-}$ $\quad$ $\frac{+}{++}$

$\ominus \quad \oplus \quad \ominus \quad \oplus$

$-6 \quad 0 \quad 2$

und $\quad$ und

$$\boxed{(-\infty, -6] \cup (0, 2)}$$

Checkpoint: Lecture 18, problem 3

Applications

ex. Find the domain of the function:

$$f(x) = \sqrt{\frac{x^2 - 2x - 8}{x^2 + 1}}$$

$$\frac{x^2 - 2x - 8}{x^2 + 1} \geq 0$$ and $x^2 + 1 \neq 0$ no real solution.

$$\frac{(x-4)(x+2)}{x^2+1} \geq 0$$

and $x^2 + 1 \neq 0$

$x = -3$	$x = 0$	$x = 5$
$- \cdot -$	$- \cdot +$	$+ \cdot +$
$+$	$-$	$+$

-2 $\quad$ 4

$$\boxed{(-\infty, -2] \cup [4, \infty)}$$

If ft/sec. For different units of measurement different a.

If an object is subject only to the force of gravity, then the object's height, $h(t)$, is given by the formula

$$h(t) = -16t^2 + v_0 t + h_0$$

384 0

where h_0 is the **initial height** in feet and v_0 is the **initial velocity** in feet per second.

NOTE: If the object is moving upward at $t = 0$, then $v_0 > 0$, whereas if the object is moving downward at $t = 0$, then $v_0 < 0$.

ex. A projectile is fired straight upward from the ground level with an initial velocity of 384 feet per second. During what time period will its height exceed 2048 feet?

① Find equation for height.

$h(t) = -16t^2 + 384t$

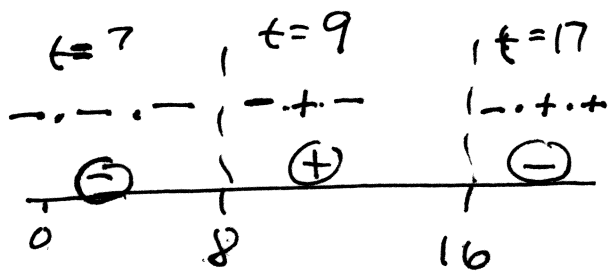
$-16t^2 + 384t > 2048$

$-16t^2 + 384t - 2048 > 0$

$-16(t^2 - 24t + 128) > 0$

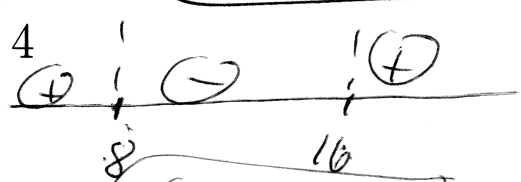
option 2 $[-16(t-8)(t-16) > 0](-\frac{1}{16})$

$(t-8)(t-16) < 0$



$8 < t < 16$ seconds

(8,16) seconds



(8,16) seconds

Checkpoint: Lecture 18, problem 4

Practice.

1) Solve the inequality: $\dfrac{1}{x^2+8x+12} \geq \dfrac{1}{x}$

2) The demand function for a product is $p(x) = -x + 10,\ 0 \leq x \leq 10$. The cost function for the product is $C(x) = 2x + 12$. Find all values of x for which the company will earn a profit. Express your answer in interval notation.

Profit = Revenue - Cost

Revenue = price · x

$R = (-x+10)x = -x^2+10x$

$P = (-x^2+10x) - (2x+12)$

$P = -x^2 + 8x - 12 > 0$

$x^2 - 8x + 12 < 0$

$(x-2)(x-6) < 0$

$+ \quad 2 \quad - \quad 6 \quad +$

$2 < x < 6$

Answer. 1) $(-\infty, -6) \cup [-4, -3] \cup (-2, 0)$ 2) $(2, 6)$